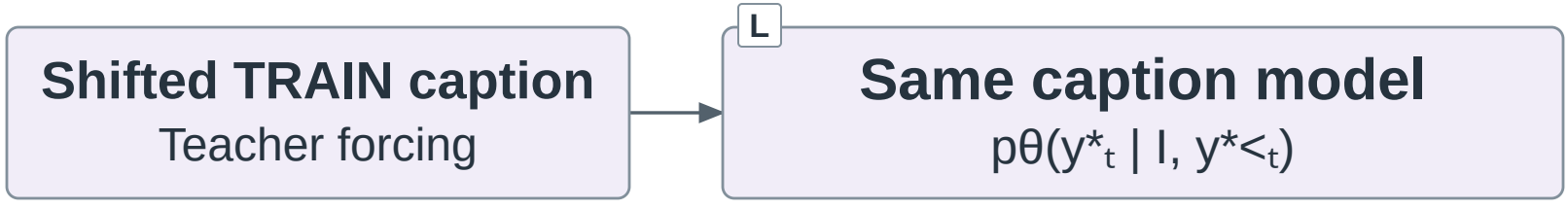
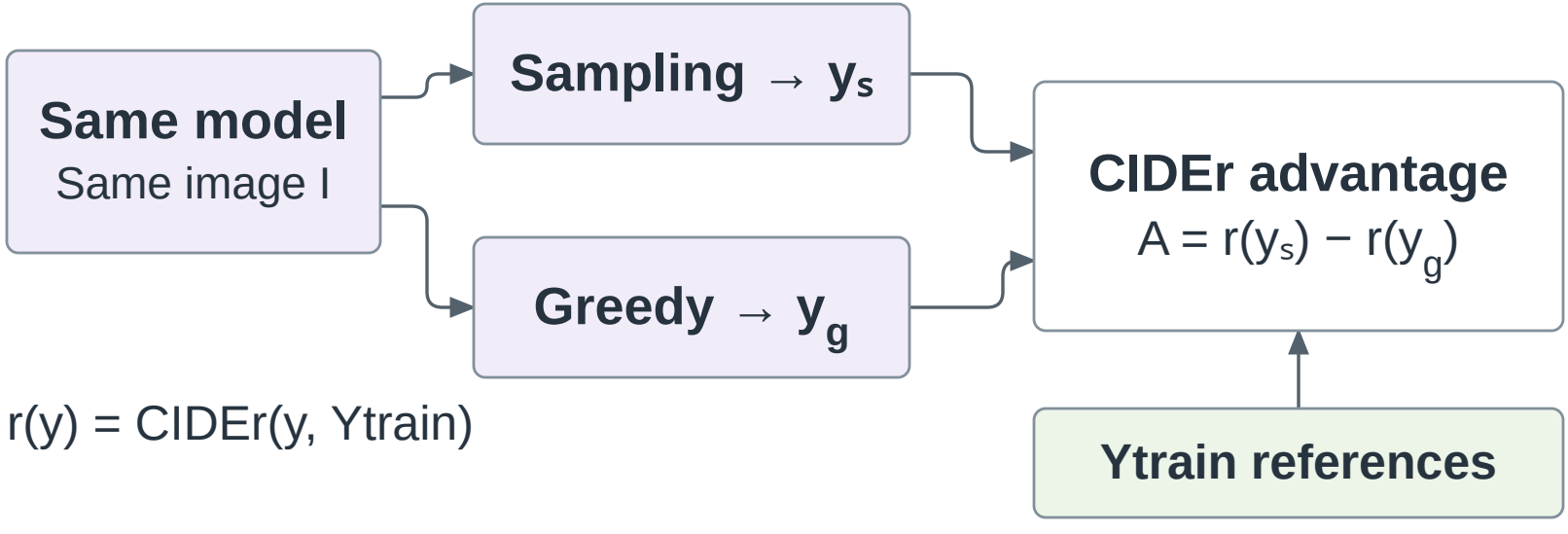
(c) Training objective

Stage 1 · Cross-entropy



$$L_{CE} = -\text{mean} \log p_\theta(y^*_t \mid I, y^*_{<t})$$

Stage 2 · Self-critical training



$$L_{SCST} = -\text{mean_batch}\,[\text{stopgrad}(A) \times \Sigma_t \log p_\theta(y_{s,t} \mid I, y_{s,<t})]$$

$$\mathbf{L = L_{CE} + 0.05\, L_{SCST}}$$